

Class 12: Guided Lab — EDA Practice

Data-driven Systems Engineering

Agenda

- Lab format: hands-on, not lecture — you drive, we circulate.
- Reusing the Class 10 EDA workflow on a dataset you have not seen yet.
- Three dataset options, same checklist, different data.
- What a finished lab notebook should contain.

Learning goals

- Apply the Class 10 EDA workflow to a new dataset without step-by-step guidance.
- Load a scikit-learn bundled dataset into a pandas DataFrame from scratch.
- Produce missing-value checks, summary statistics, and at least two plots unaided.
- State one hypothesis about feature separability, backed by what you plotted.

Course Map and Running Cases



Today in the course

Focus: Guided practice: the same EDA workflow, a dataset you pick yourself.

Bridge: Class 10 walked through EDA on iris step by step; today the steps are the same, but you choose the dataset and drive.

Fraud case

This is the same discipline the fraud case needed before any model was trusted: load, check, visualize, hypothesize.

Pasteurization case

It is also what the pasteurization case needs the first time a new batch of synthetic sensor data is generated.

Course thread: The workflow, not the dataset, is the thing you are practicing today.

How this lab is structured

1. Pick one dataset: Wine, Breast Cancer, or Digits (all built into scikit-learn).
2. Load it into a pandas DataFrame.
3. Run the same checklist as 'Class2_EDA.ipynb': missingness, descriptive statistics, distributions, correlations.
4. Write one or two hypotheses about which features separate the classes best.

Repo activity

'notebooks/Class2_EDA_Exercise.ipynb' is this lab, already in the repository — open it and work directly inside it.

Starter code

```
from sklearn.datasets import load_wine
import pandas as pd

wine = load_wine()
df_wine = pd.DataFrame(wine.data, columns=wine.feature_names)
df_wine['target'] = wine.target

print(df_wine.head())
print(df_wine['target'].value_counts())
```

- From 'Class2_EDA_Exercise.ipynb': the same pattern applies to `load_breast_cancer` and `load_digits`, just change the loader.
- Everything after this point is exactly Class 9 and Class 10 material, applied to unfamiliar columns.

Option A: Wine

- 178 samples, 13 chemical-property features, 3 classes.
- Suggested angle: how do the three wine classes differ in alcohol content, phenols, and color intensity?

Option B: Breast Cancer

- 569 samples, 30 features describing cell-nuclei characteristics, 2 classes.
- Suggested angle: which features look most discriminative between malignant and benign cases?

- 1,797 samples, 8×8 pixel images of handwritten digits, 10 classes.
- Suggested angle: what do pixel-intensity correlations and per-digit distributions look like?

The checklist, whichever dataset you picked

Step	What to produce
Missingness	<code>isnull().sum()</code> per column.
Descriptive stats	<code>.describe()</code> , and <code>.value_counts()</code> on the target.
Distributions	At least one histogram or boxplot per class.
Correlations	A correlation heatmap across numeric features.
Pairwise view	A pairplot, if the feature count makes it readable.

What a finished lab notebook should contain

- Every checklist item above, run and visible in the notebook output.
- One or two written hypotheses about which features separate the classes, in your own words.
- Nothing about modeling or metrics yet — that is deliberately still ahead of us in the Data-Driven block.

Question

Compare notes with someone who picked a different dataset. Did the same checklist surface different *kinds* of insight on Wine versus Digits?

- Makes the point that the workflow generalizes even when the data semantics are completely different.

Papers and materials

scikit-learn documentation; pandas documentation; Tukey (1977), Exploratory Data Analysis.

From This Class to the Next

What stays with us

- The EDA workflow from Class 10 transfers to any tabular dataset with minimal changes.
- Running the checklist unaided, on data you have not seen before, is the real test of whether it was learned.
- This closes the Python block: twelve classes from “what is Python” to a working, repeatable analysis habit.

Project use

Your own course project's first deliverable should look exactly like this lab's checklist, applied to your project's dataset.

Next connection

The Data-Driven block picks up from here with PCA, feature engineering, modeling, and evaluation — the theory behind the choices this lab only hinted at.

Course thread: Twelve classes of programming instrumentation now support everything the rest of the course will ask of that data.